Equations for Étendue

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1 Symbols

- α is the brightness of the target being sought in units commensurate with photons/second/meter^2
- β is the brightness of the sky (photons/second/meter^2/steradian)
- γ will be the SNR ratio
- Ω is the area of the FOV of the sensor/optical system (steradians^2)
- ω is the area of the FOV of a pixel (steradians^2) or the seeing disk or the diffraction disk, whichever is largest. Basically this is the area over which the light is spread and needs to compete with the background.
- A is the effective collecting area (meter^2).
- T is the integration time (seconds.) This may be the result of processing n shorter integrations t.
- g is used in some places as the goal limiting magitude

- q is the net quantum efficiency of the whole system
- f is the fraction of light from an object captured in the photometry aperture
- C gives the number of photons per second per squar meter for the band being used.

2 Derivation of Search Rate

The signal from an object is $S \propto \alpha A T$. Assuming we are background limited, the noise is $N \propto \sqrt{\beta A T \omega}$. Putting things together

$$SNR^2 = \gamma^2 \propto \frac{\alpha^2 A^2 T^2}{\beta A T \omega} = \frac{\alpha^2 A T}{\beta \omega} \quad \text{so} \quad \frac{1}{T} \propto \frac{\alpha^2 A}{\gamma^2 \beta \omega}.$$

This is the rate at which fields are interrogated for objects to the desired 'limiting magnitude." To get the search rate, multiply by the total area surveyed:

$$\boxed{SR = \frac{\Omega}{T} \propto \frac{\alpha^2 A \Omega}{\gamma^2 \beta \omega}} \tag{1}$$

which has units commensurate with steradians/second.

3 Integration Time Required for an SNR

Given a few things I want to find the integration time needed for a given SNR, assuming for the moment BLIP.

Factors we have in the code: g is our goal limiting magnitude.

$$S = qfAT \text{ amag}\,(g)\, C$$

$$N = \sqrt{qAt \text{ amag}\,(\beta)\, C\omega}$$

$$SNR^2 = \frac{q^2 f^2 A^2 t^2 \text{ amag}\,(g)^2\, C^2}{qAt \text{ amag}\,(\beta)\, C\omega}$$

$$\gamma^2 = \frac{qf^2 At \ \mathrm{amag}\left(g\right)^2 C}{\mathrm{amag}\left(\beta\right)\omega}$$

solving for t

$$\boxed{T = \frac{\gamma^2 \ \mathrm{amag}\left(\beta\right)\omega}{qf^2 A \ \mathrm{amag}\left(g\right)^2 C}} \tag{2}$$